

Link Layer Notes

Link Layer — Notes

1. Design Issues in the Link Layer

a) Framing

Breaks the raw bit stream from the physical layer into discrete frames. - **Byte counting** — length field indicates frame size - **Byte stuffing** — flag bytes + escape characters - **Bit stuffing** — flag bit pattern (e.g., HDLC uses 0111110) - **Physical layer coding violations** — used in some LANs

b) Error Control

Detects (and sometimes corrects) transmission errors. - **Detection**: parity bits, checksums, **CRC (Cyclic Redundancy Check)** - **Correction**: Hamming codes (used where retransmission is costly, e.g., wireless/satellite) - **ARQ (Automatic Repeat reQuest)**: acknowledgments + retransmission

c) Flow Control

Prevents a fast sender from overwhelming a slow receiver. - **Stop-and-Wait** — one frame at a time, wait for ack - **Sliding Window** (Go-Back-N, Selective Repeat) — multiple frames in flight

d) Medium Access Control (MAC)

Decides who transmits when multiple devices share a channel. - **Channel partitioning**: TDMA, FDMA, CDMA - **Random access**: ALOHA, CSMA, CSMA/CD, CSMA/CA - **Controlled access**: Token passing, polling

e) Addressing

- Uses **MAC address** (48-bit, burned into NIC)
- Distinct from IP address (network layer)

f) Link Management / Reliability

- Establishing, maintaining, terminating a logical link between directly connected nodes
 - Relevant in point-to-point protocols like **PPP**
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2. Example Protocols

Ethernet (IEEE 802.3)

- Dominant **wired LAN** technology
- Originally used **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) over shared coax/hubs
- Modern Ethernet uses **switches** → dedicated collision-free links, so CSMA/CD is largely obsolete in practice (still exam-relevant)
- **Frame format**: Preamble → Destination MAC → Source MAC → Type/Length → Data (46-1500 bytes) → FCS (CRC)
- Speed evolution: 10 Mbps → Fast Ethernet (100 Mbps) → Gigabit → 10G/40G/100G

WLAN / Wi-Fi (IEEE 802.11)

- **Wireless LAN** standard
- Uses **CSMA/CA** (Collision Avoidance), not CD — a wireless sender can't reliably detect collisions while transmitting (own signal drowns out the channel)
- **RTS/CTS** (Request to Send / Clear to Send) — solves the **hidden terminal problem** (two senders can't hear each other but both reach the same receiver)
- **Exposed terminal problem** — node unnecessarily holds back from transmitting
- **NAV (Network Allocation Vector)** — virtual carrier sensing
- **Architecture**: BSS (Basic Service Set) with Access Point, or ad hoc (no AP)
- **Frame types**: Management, Control, Data

Bluetooth (IEEE 802.15.1)

- Short-range **Wireless PAN**
- Uses **FHSS (Frequency-Hopping Spread Spectrum)** — hops ~1600 times/sec across 79 channels (2.4 GHz band)

- **Piconet:** 1 master + up to 7 active slaves
- **Scatternet:** multiple piconets linked together
- Low power, low range (~10m), used for pairing (headsets, keyboards, IoT)

3. Quick Comparison Table

Feature	Ethernet	WLAN	Bluetooth
Medium	Wired	Wireless	Wireless
Access method	CSMA/CD (legacy)	CSMA/CA	FHSS + master-slave polling
Range	Local (cable length)	~50-100m	~10m
Topology	Star (switched)	BSS / Infrastructure	Piconet / Scatternet

4. Key Exam Points to Remember

- CSMA/CD vs CSMA/CA: wired can detect collisions, wireless can only avoid them
- Hidden terminal problem → solved by RTS/CTS
- MAC address = 48-bit, Layer 2; IP address = Layer 3
- Ethernet switches removed the need for CSMA/CD in modern networks
- Bluetooth = PAN, WLAN = LAN — different scope despite both being wireless